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SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Case study

19.1.18

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization

plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems

3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
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Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

MARK ALLOCATION

	Understanding of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)
Q1	2	2	2	1.5	1
Q2	2	2	2	1.5	1
Q3	1.5	2.5	1.5	2	0.5
Q4	2.5	1.5	1.5	2	0.5
Q5	1.5	1	2	1	1
					39.5

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1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The problem is mainly caused by poor network conditions and the transport protocol behavior. The network has 8% packet loss, 180 ms jitter, and fluctuating bandwidth.

If TCP is used, lost packets are retransmitted. This improves reliability, but retransmission increases delay and may cause frozen video and delayed communication.

UDP is more suitable for real-time video and voice because it sends data faster and does not wait for retransmission.

The application layer can also affect performance. If the jitter buffer is not properly configured, irregular packet arrival can cause voice distortion and video freezing.

Justify your answer based on the communication requirements of real-time multimedia applications.

- Low latency
- Low jitter
- Continuous communication
- Adaptive bandwidth
- Fast delivery of voice and video

In video conferencing, receiving data quickly is often more important than receiving every packet.

TCP provides reliable delivery, but retransmission can increase delay.

UDP is more suitable because it provides faster communication with less delay.

Therefore, real-time applications usually prefer low latency over perfect reliability.

3. Recommend suitable protocol-level or application-level improvements.

The following improvements can improve conferencing quality:

1. Use UDP for real-time voice and video traffic.
2. Apply Quality of Service (QoS) to give priority to voice and video.
3. Use adaptive bitrate control when bandwidth changes.
4. Use a jitter buffer to reduce the effect of irregular packet arrival.
5. Use packet loss concealment to handle missing packets.
6. Reduce video quality or frame rate during network congestion.

Conclusion:

Using UDP with QoS, adaptive bitrate control, packet loss handling, and jitter buffering can improve video conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

HIGH DNS RESOLUTION TIME

1. Analyze the reasons behind the high DNS resolution time.

The DNS lookup time is approximately 800 ms. Possible reasons include:

1. Geographical distance between users and DNS servers.
2. Centralized DNS servers handling too many requests.
3. Multiple DNS lookup steps.
4. Poor DNS caching.
5. DNS server overload.
6. Network congestion or packet loss.

These factors can increase DNS resolution time and delay application loading.

2. Propose a suitable DNS architecture using Anycast DNS, DNS pre-fetching, and TTL optimization.

A distributed DNS architecture can be used.

The architecture should include:

1. Anycast DNS
Multiple DNS servers are placed in different geographical locations. Users are directed to the nearest DNS server.
2. DNS Pre-fetching
Frequently required DNS records are retrieved in advance.
3. TTL Optimization
TTL should be balanced to provide good caching while allowing DNS records to be updated when required.
4. Global Load Balancing
DNS requests should be distributed across multiple servers.

Conclusion:

Anycast DNS, caching, DNS pre-fetching, and optimized TTL can reduce DNS lookup time and improve performance.

3. Evaluate the advantages and limitations of the proposed architecture.

Advantages:

1. Lower DNS lookup latency.
2. Better performance for global users.
3. High scalability.
4. High availability.
5. Better load distribution.
6. Improved fault tolerance.

Limitations:

1. Complex configuration and management.
2. Higher infrastructure cost.
3. High TTL can delay DNS updates.
4. Low TTL can increase DNS traffic.
5. Anycast depends on proper network routing.

Conclusion:

The distributed DNS architecture provides better performance, scalability, and availability, but requires proper management.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

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5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

CENTRALIZED AND DISTRIBUTED DNS

4. Examine the impact of centralized DNS architecture on application performance.

A centralized DNS architecture sends all DNS requests to one main server.

This can cause the following problems:

1. High latency for users located far from the server.
2. Server overload during high traffic.
3. Single point of failure.
4. Limited scalability.
5. Slow application loading.

During promotional events, a large number of users send DNS requests at the same time. The single DNS server becomes overloaded, causing DNS lookup delays.

Conclusion:

Centralized DNS creates a performance bottleneck and reduces scalability and availability.

5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

A distributed DNS solution should use multiple DNS servers in different geographical locations.

The design should include:

1. Regional DNS servers in Asia, Europe, America, and other regions.
2. Anycast routing to direct users to the nearest DNS server.
3. Multiple DNS servers in each region for redundancy.
4. Load balancing to distribute traffic.
5. Health monitoring to detect failed servers.

6. Automatic failover to redirect traffic when a server fails.

Conclusion:

A distributed DNS system reduces latency, improves availability, and prevents a single point of failure.

6. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.

TTL Values

TTL controls how long DNS information remains in the cache.

High TTL:

- Reduces DNS queries.
- Reduces server load.
- Improves repeated access speed.
- Makes DNS updates slower.

Low TTL:

- Allows faster DNS updates.
- Supports faster failover.
- Increases DNS queries and server load.

Anycast Routing

Anycast allows multiple DNS servers to use the same IP address.

Benefits:

- Directs users to nearby servers.
- Reduces latency.
- Distributes traffic.
- Improves availability.
- Provides fault tolerance.

DNS Caching

DNS caching stores previous DNS responses.

Benefits:

- Faster DNS responses.
- Reduced DNS traffic.
- Lower load on DNS servers.

Limitation:

Cached records may become outdated until the TTL expires.

Conclusion:

TTL, Anycast, and DNS caching work together to improve performance, scalability, availability, and fault tolerance.

7. A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

TCP LATENCY IN STOCK TRADING

1. Analyze three possible TCP-related factors responsible for the latency increase.

The three possible factors are:

1. Flow Control and Excessive Buffering

TCP flow control prevents the sender from sending data faster than the receiver can process.

During high traffic, buffers can become full. The receiver reduces its receive window, causing the sender to slow down.

This increases transaction delay.

2. Nagle's Algorithm

Nagle's algorithm combines small packets before sending them.

This reduces the number of packets but can delay small transactions.

In stock trading, even a small delay can affect transaction performance.

3. SYN Queue Limitations

The SYN queue stores incoming TCP connection requests.

During market opening, many users may try to connect at the same time.

If the SYN queue becomes full:

- Connection requests are delayed.
- Some requests may be dropped.
- Clients may retransmit SYN packets.
- Connection establishment becomes slower.

2. Explain how each factor affects transaction processing in high-frequency systems.

Flow Control

When buffers become full, the sender slows down.

Effects:

- Orders wait in buffers.
- Acknowledgments are delayed.
- Transaction processing becomes slower.

Nagle's Algorithm

Small packets may wait before transmission.

Effects:

- Buy and sell orders are delayed.
- Trade confirmations become slower.
- System response time increases.

SYN Queue Limitations

When the SYN queue is full, new connections are delayed.

Effects:

- Connection establishment takes more time.
- More retransmissions occur.
- Order submission is delayed.

Conclusion:

These factors can increase latency from 1 ms to 40 ms and reduce the performance of the trading system.

3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

1. Optimize TCP send and receive buffer sizes.
2. Disable Nagle's algorithm using TCP_NODELAY for latency-sensitive transactions.
3. Increase the SYN backlog or SYN queue capacity.
4. Use persistent TCP connections instead of creating a new connection for every transaction.
5. Monitor and reduce TCP retransmissions.
6. Reduce packet loss and network congestion.
7. Apply traffic prioritization for important trading traffic.

Simple Summary:

Optimize TCP Buffers
+
Enable TCP_NODELAY
+
Increase SYN Queue
+
Use Persistent Connections
+
Reduce Packet Loss
=
Lower Latency + Better Reliability

Conclusion:

By optimizing TCP buffers, disabling unnecessary packet delays, increasing SYN queue capacity, using persistent connections, and reducing packet loss, the system can achieve lower latency and better transaction reliability.

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